APPENDIX E

Glossary

Working with Biological Data in Python and R

A consolidated reference to the key terms used throughout the book — programming, statistics, sequences and genomics, RNA-seq, ecology, phylogenetics, machine learning, and reproducible workflows. Each entry notes the chapter where the term is introduced or used most. Terms in *italics* within a definition have their own entry.

A

Accuracy — The fraction of predictions a classifier gets right. A blunt summary that can mislead with imbalanced classes — read alongside the confusion matrix. **[Ch 20]**

Aesthetic mapping — In ggplot2, the link between a variable and a visual property (x, y, colour, shape) declared inside aes(). **[Ch 8]**

Allometry — How one body measurement scales with another, usually a straight line on log–log axes; the slope is the scaling exponent. **[Ch 6, 12]**

Alpha diversity — The diversity within a single site or sample (richness plus evenness). **[Ch 17]**

Alternative hypothesis — The claim that there is a real effect or difference — what you accept if the null hypothesis is rejected. **[Ch 11]**

AnnData (.h5ad) — A container pairing an expression matrix with aligned cell and gene metadata; the standard object for single-cell analysis in Python (Scanpy). **[Ch 21]**

ANOVA — Analysis of variance: a test for a difference in means across three or more groups. **[Ch 11]**

Argument — A value passed into a function to control what it does, e.g. na.rm = TRUE. **[Ch 2]**

Assumption (of a test) — A condition a statistical test relies on (e.g. normality, equal variance); violating it can invalidate the result. **[Ch 11, 13]**

B

Base (nucleotide) — One of the letters of DNA (A, C, G, T) or RNA (A, C, G, U). **[Ch 14]**

Batch effect — Unwanted variation from samples being processed at different times or places; a model may learn it instead of the biology. **[Ch 13, 20]**

Bayesian inference — Inference that produces a distribution of plausible answers (a posterior); in phylogenetics it gives posterior-probability support for clades. **[Ch 18]**

BED — A plain-text format listing genomic intervals (chromosome, start, end) using 0-based, half-open coordinates. **[Ch 15]**

Benjamini–Hochberg (BH) — A procedure that controls the false discovery rate when many tests are run at once; the default multiple-testing correction in genomics. **[Ch 13, 16]**

Bernoulli distribution — The distribution of a single yes/no (0/1) outcome with a fixed probability. **[Ch 10]**

Beta diversity — How much community composition differs between sites. **[Ch 17]**

Bioconductor — A curated R project of 2,400+ interoperable packages for biological data; installed with BiocManager. **[Ch 21]**

Biopython — A Python library for biological sequence and file manipulation. **[Ch 14]**

Boolean — A value that is either TRUE or FALSE (True/False in Python). **[Ch 1]**

Bootstrap — Resampling the data with replacement to gauge confidence; in phylogenetics, the percentage of resampled trees that recover a given clade. **[Ch 18]**

Boxplot — A plot summarising a distribution by its median, quartiles, and outliers. **[Ch 9, 11]**

Branch (tree) — An edge of a phylogenetic tree; its length usually represents evolutionary change or time. **[Ch 18]**

Branch length — The amount of change or elapsed time along a branch of a phylogenetic tree. **[Ch 18]**

Bray–Curtis dissimilarity — A measure of how different two communities are based on their species abundances (0 = identical, 1 = no species shared). **[Ch 17]**

C

Cell (notebook) — A unit of a Jupyter notebook holding either Markdown text or code. **[Ch 19]**

Clade — An ancestor together with all of its descendants — a monophyletic group; the meaningful unit on a phylogenetic tree. **[Ch 18]**

Cladogram — A tree drawn to show only branching order, ignoring branch lengths. **[Ch 18]**

Classification — A supervised-learning task that predicts a category (e.g. tissue type) for each sample. **[Ch 20]**

Cloud computing — Renting on-demand compute (AWS, Google Cloud, Azure), paying by the hour, to scale beyond a laptop. **[Ch 21]**

Codon — A triplet of nucleotides that specifies one amino acid (or a stop signal). **[Ch 14]**

Commit — A saved snapshot of a project's files in Git, recorded with a message. **[Ch 19]**

Confidence interval — A range of plausible values for a quantity, consistent with the data at a stated confidence level (e.g. 95%). **[Ch 10, 11]**

Confusion matrix — A table of true versus predicted classes; its diagonal is correct predictions, off-diagonal cells are specific errors. **[Ch 20]**

Container — A package of code plus its entire software environment (e.g. Docker) so an analysis runs identically anywhere. **[Ch 21]**

Continuous variable — A measurement that can take any value in a range (e.g. body length), as opposed to a count or category. **[Ch 10]**

Correlation — A number from −1 to 1 measuring how strongly two variables move together; it does not imply causation. **[Ch 6]**

Count matrix — A table of read counts with genes in rows and samples in columns — the starting point of RNA-seq analysis. **[Ch 16]**

CRAM — A reference-based, highly compressed format for sequence alignments. **[Ch 15]**

Cross-validation — Rotating which fold of the data is held out for testing so every sample is tested once; gives a stable, honest performance estimate. **[Ch 20]**

CSV — Comma-separated values: a plain-text table, the universal exchange format for tabular data. **[Ch 3]**

D

Data cleaning — Turning raw, messy data into a consistent, analysable form: fixing types, missing values, duplicates, and inconsistent labels. **[Ch 5]**

Data frame — A table of data with named columns of possibly different types — the central object for tabular analysis in R and pandas. **[Ch 4]**

Data leakage — Any way test-set information seeps into training (e.g. scaling on all the data); it inflates the reported score. **[Ch 20]**

Decision boundary — The surface in feature space where a classifier switches from predicting one class to another. **[Ch 20]**

Delimiter — The character separating fields in a text table (comma, tab, semicolon). **[Ch 3]**

Dependent variable — The outcome a model tries to explain or predict (the response, y). **[Ch 12]**

DESeq2 — A Bioconductor package for differential-expression analysis of RNA-seq counts. **[Ch 16]**

Differential expression — Testing which genes change in expression between conditions. **[Ch 16]**

Dispersion — How much more variable RNA-seq counts are than a simple Poisson model predicts; estimated per gene before testing. **[Ch 16]**

Distribution — The pattern of values a variable takes — e.g. normal, log-normal, Poisson, Bernoulli. **[Ch 10]**

DOI — A permanent, citable identifier for a dataset or software release (minted by services like Zenodo). **[Ch 21]**

E

Effect size — The magnitude of a difference or relationship (e.g. a fold change), distinct from its statistical significance. **[Ch 11, 13]**

Evenness — How equally individuals are spread across species; high evenness means no single species dominates. **[Ch 17]**

Exon — A segment of a gene retained in the mature mRNA. **[Ch 15]**

F

Factor — R's data type for a categorical variable, with defined levels. **[Ch 4]**

FAIR — A guiding ideal for research outputs: Findable, Accessible, Interoperable, Reusable. **[Ch 21]**

False discovery rate (FDR) — The expected fraction of declared discoveries that are actually false; controlled by the Benjamini–Hochberg procedure. **[Ch 13, 16]**

FASTA — A plain-text format for nucleotide or protein sequences, each preceded by a ">" header line. **[Ch 14]**

FASTQ — A text format storing sequencing reads together with a per-base quality score. **[Ch 14]**

Feature — An input measurement a model uses to make predictions (a column of X). **[Ch 20]**

Feature importance — A ranking of how much each feature contributes to a model's predictions. **[Ch 20]**

Feature scaling — Putting features on a common scale (e.g. mean 0, SD 1) so magnitude differences don't bias an algorithm. **[Ch 20]**

Floating-point number — A computer's approximate representation of a real number; exact equality tests on them often fail. **[Appendix]**

Fold change — The ratio of expression between two conditions, usually reported as log2 fold change. **[Ch 16]**

Function — A reusable, named block of code that takes inputs and returns a result. **[Ch 2]**

G

Gamma diversity — The total diversity of a whole region across all its sites. **[Ch 17]**

GC content — The fraction of a sequence that is G or C. **[Ch 14]**

GenBank — A richly annotated sequence file format from NCBI. **[Ch 15]**

Gene — A stretch of DNA encoding a product (often a protein); on a tree or matrix, a unit of analysis. **[Ch 14, 16]**

Generalised linear model (GLM) — A regression framework extending linear models to non-normal outcomes (e.g. logistic regression for 0/1 data). **[Ch 12]**

geom — A ggplot2 layer that draws data a particular way (geom_point, geom_boxplot, ...). **[Ch 8]**

GFF / GTF — Plain-text formats describing genome features (genes, exons) and their coordinates, in nine columns. **[Ch 15]**

ggplot2 — R's grammar-of-graphics plotting package, building plots from data, aesthetics, and geoms. **[Ch 8]**

Git — A version-control system that records a project's full history as a series of commits. **[Ch 19]**

GitHub — An online host for Git repositories, providing backup, sharing, and collaboration. **[Ch 19]**

.gitignore — A file listing paths Git should not track (large data, secrets, clutter). **[Ch 19]**

Group-by — Splitting data into groups to compute a summary per group (dplyr group_by, pandas groupby). **[Ch 6]**

H

HDF5 — A binary format for large, hierarchical numeric arrays; underlies .h5ad and .loom. **[Ch 21]**

Header — The first row of a data file naming the columns. **[Ch 3]**

Histogram — A plot of how often values fall into each bin — a picture of a distribution. **[Ch 9, 10]**

HPC cluster — A high-performance computing cluster: many shared machines on which jobs are scheduled (often by Slurm). **[Ch 21]**

Hypothesis test — A procedure that weighs evidence in data against a null hypothesis, yielding a p-value. **[Ch 11]**

I

Independent variable — A predictor a model uses to explain the outcome (x). **[Ch 12]**

Indexing — Selecting elements by position or name; R counts from 1 and includes range ends, Python from 0 and excludes them. **[Ch 1, Appendix]**

Interquartile range (IQR) — The spread of the middle 50% of the data (Q3 − Q1); a robust measure of variability. **[Ch 10]**

Intron — A segment of a gene removed from the mRNA during splicing. **[Ch 15]**

J

Jaccard index — A presence/absence measure of how many species two communities share. **[Ch 17]**

Jupyter notebook — A Python document (.ipynb) of Markdown and code cells that mixes prose, code, and output. **[Ch 19]**

K

k-fold cross-validation — Cross-validation that splits the data into k folds and rotates the held-out test fold k times. **[Ch 20]**

k-nearest neighbours (k-NN) — A classifier that labels a sample by the majority vote of its k closest neighbours. **[Ch 20]**

Kernel — The live process that runs a notebook's code; restart it for a clean, reproducible run. **[Ch 19]**

L

Label — The known answer a supervised model learns to predict (a value of y). **[Ch 20]**

Ladderize — Reordering a tree's clades by size for a tidy staircase layout (purely cosmetic). **[Ch 18]**

Library / package — A bundle of reusable functions you load to extend R or Python. **[Ch 2]**

Linear regression — Fitting a straight line that best predicts a continuous outcome from one or more predictors. **[Ch 12]**

Literate programming — Interleaving prose, code, and its output in one document (R Markdown, Jupyter). **[Ch 19]**

Log-normal distribution — A right-skewed distribution whose logarithm is normal; common for masses and concentrations. **[Ch 10]**

Log2 fold change — The base-2 logarithm of a fold change; +1 means a doubling, −1 a halving. **[Ch 16]**

Logistic regression — A GLM that models the probability of a binary outcome (0/1). **[Ch 12, 20]**

Long format — A tidy table with one observation per row (often reshaped from wide format). **[Ch 5, 6]**

M

MA plot — A scatter of log fold change against mean expression, used to inspect differential-expression results. **[Ch 16]**

Machine learning — Fitting models that learn patterns from data in order to predict or describe new data. **[Ch 20]**

Matplotlib — Python's foundational plotting library. **[Ch 9]**

Maximum likelihood — Choosing the model (e.g. tree) that makes the observed data most probable; the workhorse of modern phylogenetics. **[Ch 18]**

Mean — The arithmetic average of a set of numbers. **[Ch 10]**

Median — The middle value of ordered data; robust to outliers. **[Ch 10]**

Merge / join — Combining two tables on a shared key column. **[Ch 6]**

Michaelis–Menten — A nonlinear model of enzyme reaction rate versus substrate concentration (parameters Vmax and Km). **[Ch 12]**

Missing value — An absent observation, represented as NA in R or NaN/None in Python; it propagates through calculations unless handled. **[Ch 5, Appendix]**

Monophyletic — Describing a group that contains an ancestor and all of its descendants — a clade. **[Ch 18]**

Most recent common ancestor (MRCA) — The node where the lineages of a set of tips first join. **[Ch 18]**

Motif — A short recurring sequence pattern (e.g. a promoter element or binding site). **[Ch 14]**

Multiple testing — Running many hypothesis tests at once, which inflates false positives unless p-values are corrected. **[Ch 13, 16]**

N

NA / NaN / None — Markers for missing or undefined values; NA is R's missing value, NaN is "not a number", None is Python's null. **[Ch 5]**

Negative binomial — The count distribution used to model RNA-seq data because counts are more variable (overdispersed) than Poisson. **[Ch 16]**

Neighbour-joining — A fast distance-based method for building phylogenetic trees. [Ch 18]

Newick — A compact text notation for a phylogenetic tree, nesting clades in parentheses. [Ch 18]

NEXUS — A rich text container holding a tree plus its alignment and analysis commands. [Ch 18]

nf-core — A community library of standardised, peer-reviewed Nextflow pipelines. [Ch 21]

NMDS — Non-metric multidimensional scaling: an ordination that places similar communities close together in 2-D. [Ch 17]

Normal distribution — The symmetric, bell-shaped distribution; many measurements are approximately normal. [Ch 10]

Normalisation — Adjusting counts so samples are comparable despite different sequencing depths (e.g. size factors, CPM). [Ch 16]

Null hypothesis — The default claim of no effect or no difference, which a test seeks to reject. [Ch 11]

O

Open science — Making the whole research process — code, data, and findings — open and reusable. [Ch 21]

Ordination — Placing samples in a low-dimensional space so similar ones sit close (PCoA, NMDS, PCA). [Ch 17]

Outgroup — A taxon known to branch off earliest, used to root a phylogenetic tree. [Ch 18]

Outlier — A value far from the rest of the data; may be an error or a real extreme. [Ch 10]

Overdispersion — More variability than a simple model (e.g. Poisson) predicts; the reason RNA-seq uses the negative binomial. [Ch 16]

Overfitting — When a model fits the training data's noise, scoring well on it but poorly on new data. [Ch 20]

P

p-value — The probability of seeing data at least as extreme as observed if the null hypothesis were true; small values are evidence against the null. [Ch 11]

Paired test — A test for differences between two measurements on the same units (e.g. before/after). [Ch 11]

Parametric test — A test assuming the data follow a particular distribution (often normal); nonparametric tests do not. [Ch 11]

Patristic distance — The total branch length along the path between two tips on a tree — its measure of evolutionary distance. [Ch 18]

PCA — Principal component analysis: an unsupervised method that summarises many variables in a few axes of greatest variation. [Ch 16]

PCoA — Principal coordinates analysis: ordination of a distance matrix (e.g. Bray–Curtis) into 2-D. [Ch 17]

PDB / mmCIF — Formats for the 3-D atomic coordinates of macromolecular structures (mmCIF is the modern standard). [Ch 21]

PERMANOVA — A permutation-based multivariate test for whether community composition differs between groups. [Ch 17]

Phylogenetic tree — A branching diagram of evolutionary relationships among species, genes, or strains. [Ch 18]

Phylogram — A tree drawn with branch lengths to scale (showing amount of change). **[Ch 18]**

Pielou's evenness (J) — Evenness expressed on a 0–1 scale, independent of richness. **[Ch 17]**

Pipe — An operator (|> in R, method chaining in pandas) that passes a result into the next step for readable workflows. **[Ch 6]**

Pipeline (ML) — A bundle of preprocessing plus a model so steps like scaling stay inside cross-validation and don't leak. **[Ch 20]**

Pipeline (workflow) — A multi-step analysis whose steps and dependencies are managed automatically (Snakemake, Nextflow). **[Ch 21]**

Poisson distribution — The distribution of counts of independent events in a fixed interval. **[Ch 10]**

Population vs sample — The whole group of interest versus the subset actually measured; statistics infer the former from the latter. **[Ch 10]**

Precision — Of the items a classifier labelled positive, the fraction that truly are. **[Ch 20]**

Predictor — A variable used as input to a model (an independent variable / feature). **[Ch 12]**

Preprint — A manuscript shared openly (e.g. on bioRxiv) before formal peer review. **[Ch 21]**

Promoter — A DNA region upstream of a gene that controls its transcription. **[Ch 14, 15]**

Pseudoreplication — Treating sub-samples as independent replicates when the treatment was applied at a higher level (e.g. fish within a tank). **[Ch 13]**

Pull request — A proposed, reviewable set of changes on GitHub. **[Ch 19]**

Q

Quality score — A FASTQ value encoding the confidence of each sequenced base. **[Ch 14]**

Quantile — A cut point dividing ordered data into equal portions (e.g. quartiles, percentiles). **[Ch 10]**

R

R Markdown — R's literate-programming format (.Rmd), knitted to HTML/PDF/Word. **[Ch 19]**

Random forest — An ensemble classifier averaging many decision trees. **[Ch 20]**

Rarefaction — Interpolating richness down to a common sampling effort so unequal samples can be compared fairly. **[Ch 17]**

Recall — Of the truly positive items, the fraction a classifier correctly found. **[Ch 20]**

Reference level (baseline) — The factor level a model compares others against; in R it defaults to the first level alphabetically. **[Ch 12, Appendix]**

Regression — Predicting a continuous outcome from one or more predictors. **[Ch 12]**

Regularisation — A penalty added to a model to discourage over-complex fits and reduce overfitting. **[Ch 20]**

Relative path — A file path written relative to the project root, so code runs on any machine. **[Ch 19]**

Repository (repo) — A project folder whose full history Git tracks. **[Ch 19]**

Reproducible — Able to be repeated: the same data and code give the same result. **[Ch 19]**

Residual — The difference between an observed value and a model's prediction. **[Ch 12]**

Reverse complement — The complementary DNA strand read in the opposite direction. **[Ch 14]**

Richness — The number of species (or taxa) present in a sample. **[Ch 17]**

Root (tree) — The deepest common ancestor, which gives a tree direction in time. **[Ch 18]**

S

SAM / BAM — Text and binary formats for sequencing reads aligned to a reference genome. **[Ch 15]**

Scanpy — A Python toolkit for single-cell analysis built on AnnData. **[Ch 21]**

scikit-learn — Python's standard machine-learning library, with a uniform fit/predict interface. **[Ch 20]**

Seaborn — A Python library for statistical plots built on matplotlib. **[Ch 9]**

Seed (random) — A number that fixes a random-number generator so a stochastic analysis repeats exactly. **[Ch 19, 20]**

Shannon diversity (H′) — A diversity index combining richness and evenness, sensitive to rare species. **[Ch 17]**

Significance level (α) — The p-value threshold below which a result is called statistically significant (commonly 0.05). **[Ch 11]**

Simpson diversity — A diversity index weighted toward common species. **[Ch 17]**

Sister taxa — Two tips that share a most recent common ancestor with each other and no one else. **[Ch 18]**

Size factor — A per-sample scaling that normalises for sequencing depth in RNA-seq. **[Ch 16]**

Skew — Asymmetry of a distribution; right-skewed data have a long upper tail and mean > median. **[Ch 10]**

Snakemake — A Python-based workflow manager that runs pipeline rules in dependency order. **[Ch 21]**

Stage (Git) — Marking edited files (git add) to include them in the next commit. **[Ch 19]**

Standard deviation — A measure of how spread out values are around the mean. **[Ch 10]**

Standardisation (z-score) — Rescaling a variable to mean 0 and standard deviation 1. **[Ch 10, 20]**

StandardScaler — scikit-learn's tool for standardising features; fit on training data only. **[Ch 20]**

Stratify — Splitting data so each class keeps its proportion in the training and test sets. **[Ch 20]**

String — A piece of text data. **[Ch 1]**

Supervised learning — Learning to predict a known label from labelled examples. **[Ch 20]**

T

t-test — A test for a difference in means between two groups; Welch's version does not assume equal variances. **[Ch 11]**

Test set — Held-out data used once, at the end, to measure a model's true performance. **[Ch 20]**

Tidy data — A table with one variable per column and one observation per row. **[Ch 5]**

Tip (leaf) — A terminal node of a phylogenetic tree — a present-day taxon. **[Ch 18]**

Training set — The data a model learns from. **[Ch 20]**

Transcription — Copying DNA into RNA. **[Ch 14]**

Translation — Converting a coding sequence's codons into a protein's amino acids. **[Ch 14]**

Type / class — What kind of value an object is (number, string, logical, data frame, ...). **[Ch 1, Appendix]**

Type I / Type II error — A false positive (rejecting a true null) / a false negative (missing a real effect). **[Ch 11]**

U

Unsupervised learning — Finding structure in data without labels (clustering, PCA). **[Ch 20]**

UTR — An untranslated region at either end of an mRNA. **[Ch 15]**

V

Variance — The average squared deviation from the mean; the square of the standard deviation. **[Ch 10]**

VCF — A text format listing genetic variants relative to a reference, with per-sample genotypes. **[Ch 21]**

Vector — An ordered collection of values of the same type (R) — the basic data container. **[Ch 1]**

Vectorisation — Operating on whole vectors/columns at once instead of looping element by element; faster and clearer. **[Ch 1, Appendix]**

Version control — Tracking the full history of changes to a project (e.g. with Git). **[Ch 19]**

Volcano plot — A scatter of significance against fold change, summarising a differential-expression analysis. **[Ch 16]**

W

Welch's t-test — A t-test that does not assume the two groups have equal variances; R's default, and the safer choice. **[Ch 11, Appendix]**

Wide format — A table with one row per unit and repeated measurements spread across columns (the opposite of long format). **[Ch 5]**

Wilcoxon / Mann–Whitney — Nonparametric tests for differences between paired or independent groups, used when normality is doubtful. **[Ch 11]**

Workflow manager — Software (Snakemake, Nextflow) that runs a pipeline's steps in order, in parallel, re-running only what changed. **[Ch 21]**

Working directory — The folder R or Python treats as the current location for reading and writing files. **[Ch 2]**

Z

Zenodo — An archive that mints a citable DOI for a software or data release. **[Ch 21]**

209 terms. See also the R ↔ Python Cheat Sheet, Common Biology File Formats, and the Troubleshooting & Debugging guide.